Lecture 13

Query Processing and Optimization

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Query Processing

- A **database query** expressed in a **high-level query language** such as **SQL** must first be ***scanned, parsed***, and ***validated*.**
- The **scanner** identifies the **query tokens**—such as **SQL keywords, attribute names**, and **relation names** that appear in the text of the **query.**
- The **parser checks** the **query syntax** to **determine** whether it is **formulated according** to the **syntax rules** (rules of grammar) of the **query language.**
- The **query** must also be **validated** by checking that **all attribute** and **relation names** are **valid** and **semantically meaningful names** in the **schema** of the particular **database** being queried.
- An **internal representation** of the **query** is then **created,** usually as a **tree data structure** called a **query tree**.

Query Optimization

- The **internal representation** of a **query** i.e. **query tree** is now passed to the **Query Optimization** module for finding the ***optimal execution plan***.
- A **query** typically has **many possible execution strategies**.
- The **process** of **choosing** a **suitable execution strategies** for **processing a query** is known as **query optimization**.
- There are **two main techniques** that are **employed** during **query optimization:**
  - ***Heuristic based Query Optimization***
  - ***Cost based Query Optimization***

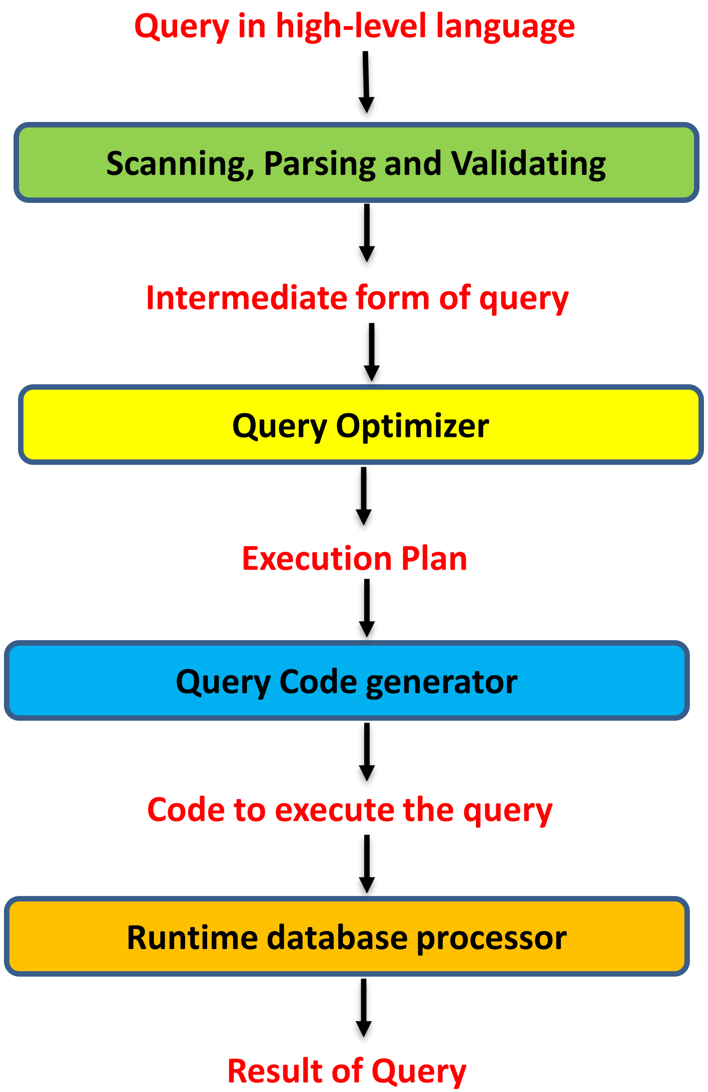
Query Optimization

- **Heuristic based Query Optimization**
  - This **technique** is based on **heuristic rules** for **ordering the operations** in a **query execution strategy.**
  - **A heuristic** is a **rule** that **works well** in **most cases** but is **not guaranteed** to **work well** in **every case.**
  - The **rules** typically **reorder the operations** in a **query tree.**

Query Optimization

- **Cost based Query Optimization**
  - This **technique** involves **systematically estimating** the **cost** of **different execution strategies :**
    - *Access cost,*
    - *Storage cost,*
    - *Computation cost,*
    - *Communication cost.*
  - And then **choosing** the **execution plan** with the **lowest cost estimate.**

Query Processing & Optimization



Query Optimization

- **Why optimizing a query?**
  - So that it is **processed efficiently.**
- **What is meant by efficiently?**
  - **Minimizing** the **cost** of **query evaluation.**
- **Who will minimize?**
  - The **system, not** the **user**.
- ***Query optimization*** *is facilitating a **system** to **construct** a **query-evaluation plan** for **processing a query efficiently, without expecting users** to **write efficient queries.***

Query Tree

- **Tree** that represents a **relational algebra expression** corresponding to a **SQL statement** is referred as **Query Tree.**
  - **Leave nodes** represent the **base tables.**
  - **Internal nodes** represent the **relational algebra operators** applied to the **child nodes.**
  - The **query tree** is **executed** from **leaves to root.**
  - An **internal node** can be **executed** when its **operands** are **available** (children relation has been computed)
  - **Internal Node** is **replaced** by the **result of the operation** it represents.
  - **Root node** is **executed last**, and is **replaced** by the **result** of the **entire tree.**

Canonical query tree

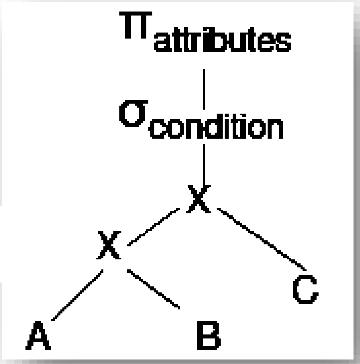
- **Construct** the **Canonical(initial) query tree** as follows:
  - **Cartesian product** of the **FROM-tables**
  - **Select** with **WHERE-condition**
  - **Project** to the **SELECT-attributes**

- **Example:**

  SELECT attributes

  FROM A, B, C

  WHERE condition



Canonical Query Tree

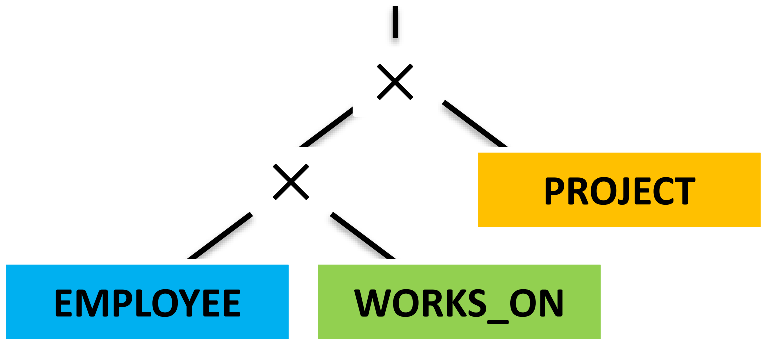
- Example: ***List the last name of the employees born after 1957 who work on a project named "Aquarius".***

SELECT *E*.LNAME **FROM** EMPLOYEE ***E*,** WORKS_ON ***W***, PROJECT ***P***

WHERE *P*.PNAME = 'Aquarius' **AND** *P*.PNUMBER = *W*.PNO **AND** *W*.ESSN = *E*.SSN **AND** *E*.BDATE > '1957-12-31'

$$\pi_{\text{LNAME}}$$

$$\sigma_{P.\text{PNAME} = \text{'Aquarius' AND } P.\text{PNUMBER} = W.\text{PNO AND } W.\text{ESSN} = E.\text{SSN AND } E.\text{BDATE} > \text{'1957-12-31'}}$$



Heuristics based Query Optimization

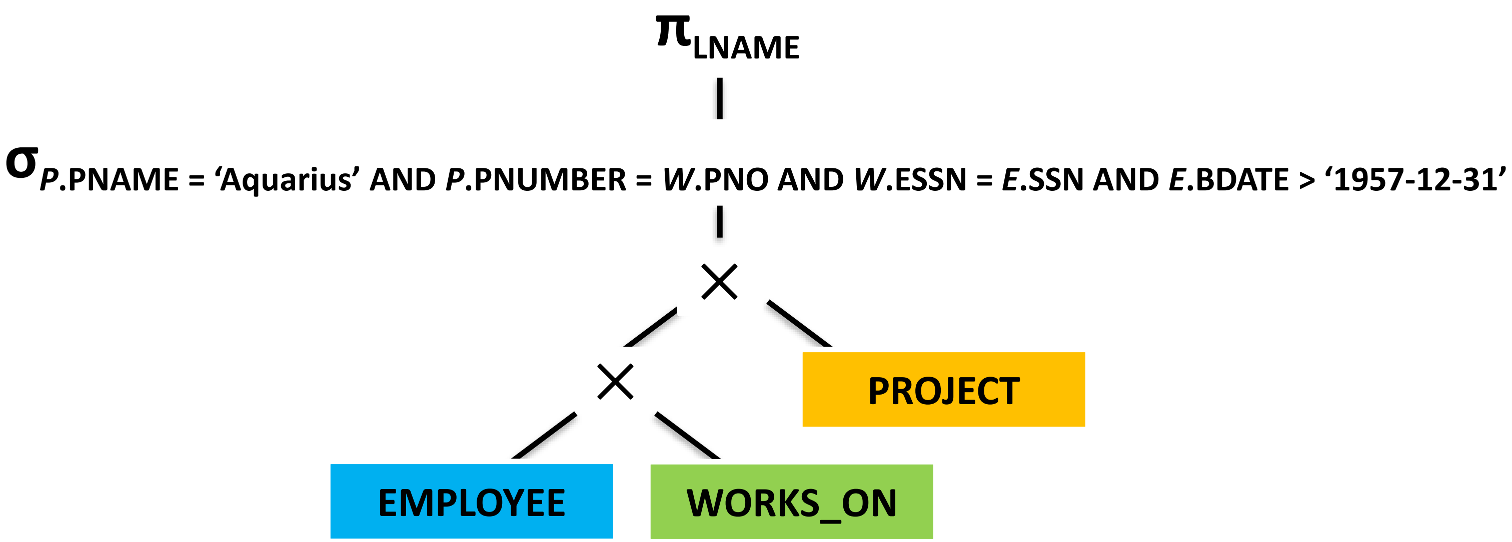
- **Heuristic rules** are used to **modify** the **internal representation** of a **query**—which is usually in the form of a **query tree** data structure—to **improve** its **expected performance.**
- The **scanner** and **parser** of an **SQL query** first generate a **data structure** that corresponds to an ***initial query representation,*** which is then **optimized** according to **heuristic rules**.
- This leads to an ***optimized query representation***, which corresponds to the **query execution strategy.**
- Following that, a **query execution plan** is **generated** to **execute** **groups** of **operations** based on the **access paths** available on the **files involved** in the **query**.

Heuristics based Query Optimization

- One of the **main heuristic rules** is :
- *Apply* ***SELECT*** *and* ***PROJECT*** *operations before applying the* ***JOIN*** or other **binary operations**.
- Because the **size** of the **file** resulting from a **binary operation**—such as **JOIN**—is usually a **multiplicative function** of the **sizes** of the **input files.**
- The **SELECT** and **PROJECT** operations **reduce the size** of a **file** and hence should be **applied before a join** or other **binary operation.**
- Thus in a **Query tree** we should **try to move SELECT and PROJECT operations down**.

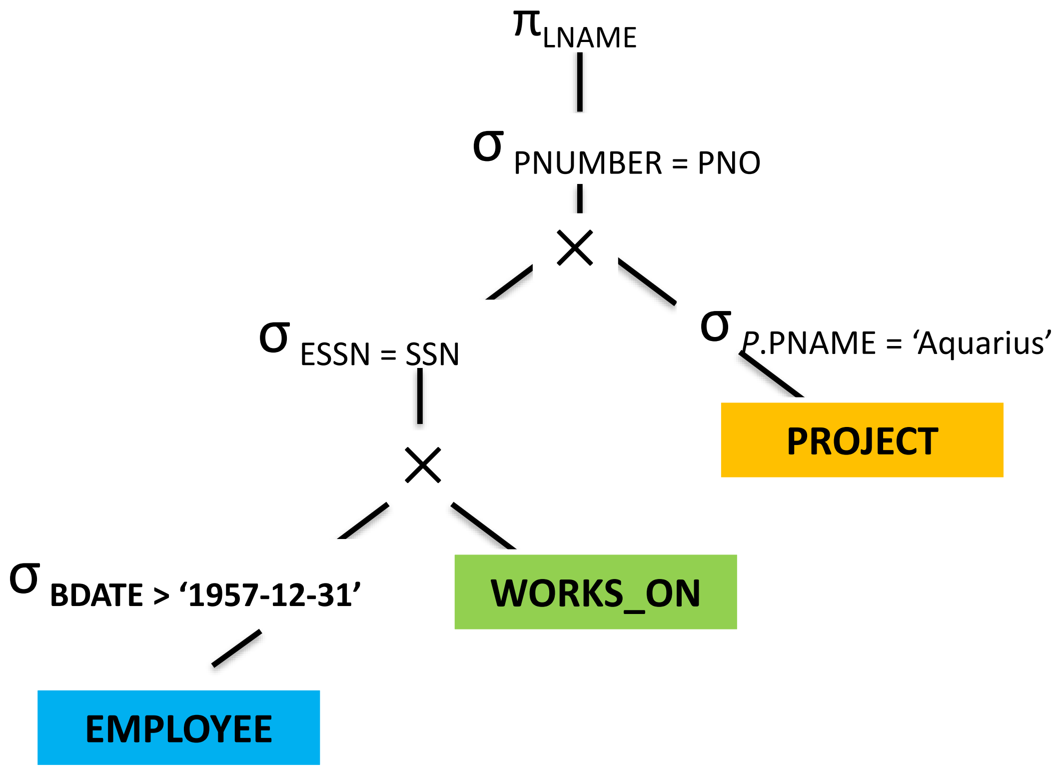
Heuristics based Query Optimization

- **Example:** *Move **SELECT** down*:



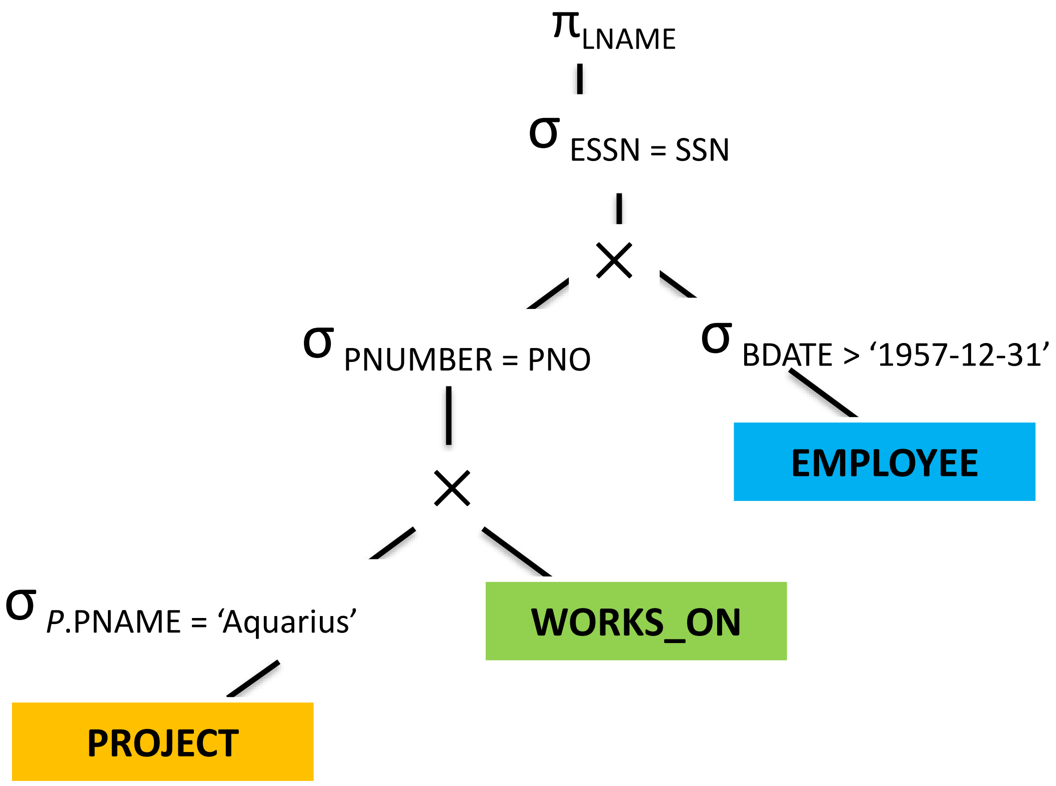
Heuristics based Query Optimization

- **Example:** Move **SELECT** **down**:



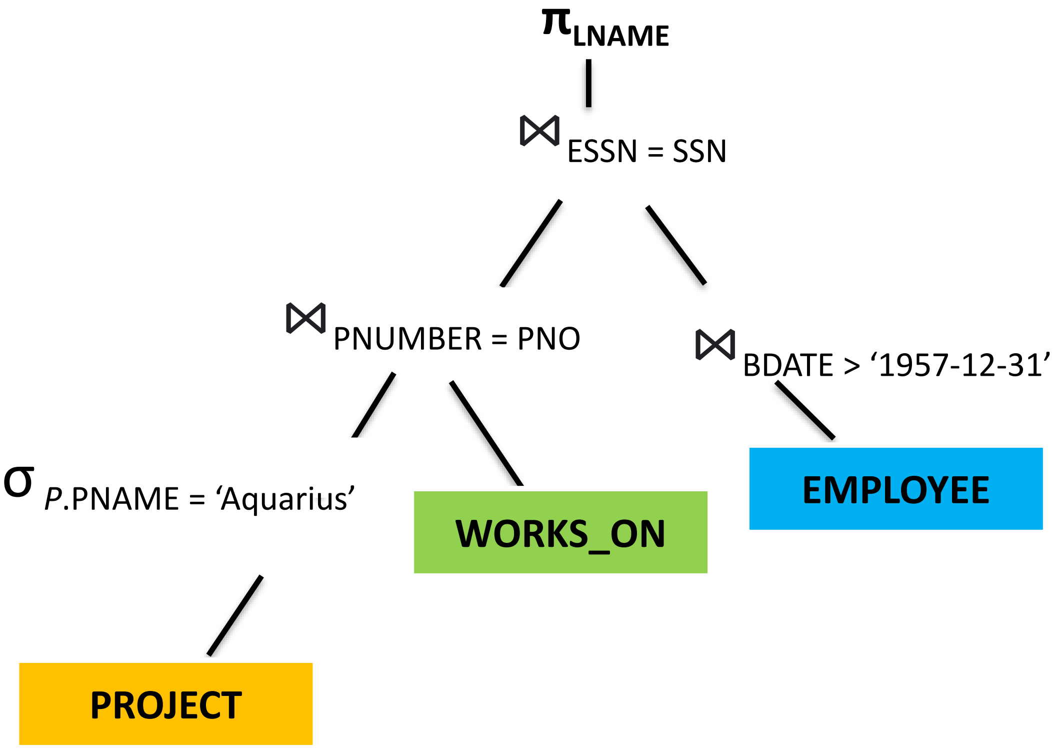
Heuristics based Query Optimization

- **Example:** Apply **most restrictive Select first.**



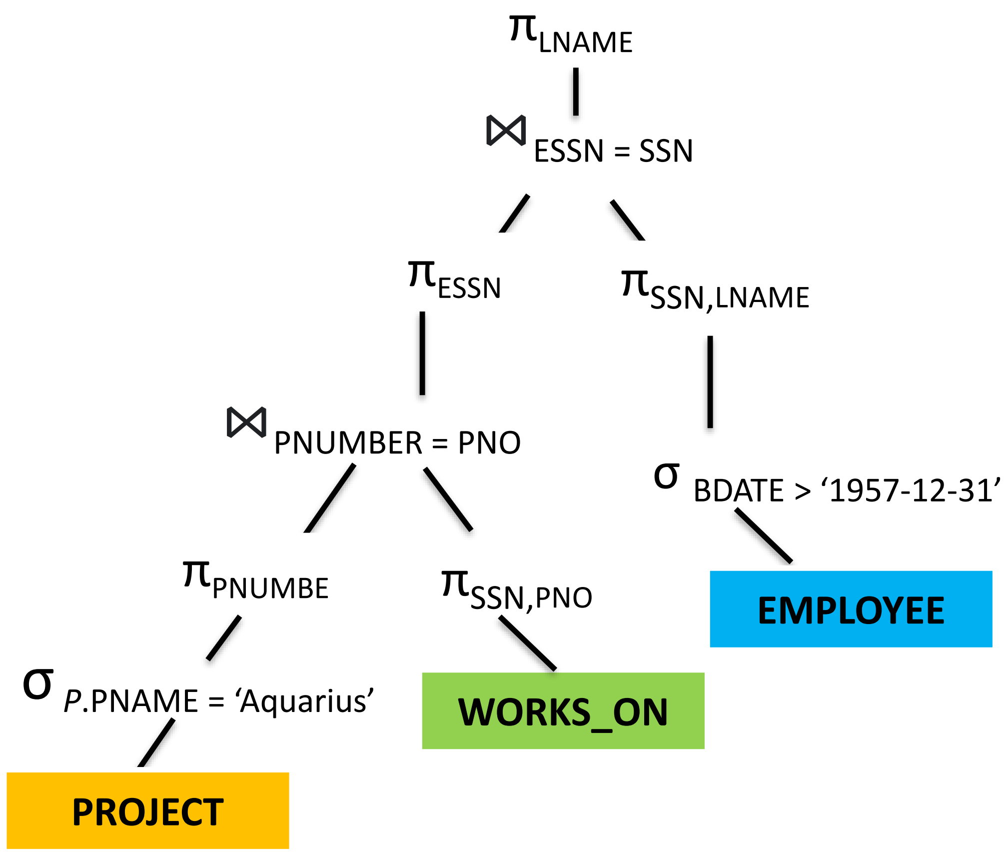
Heuristics based Query Optimization

- **Example:** Replace **Cartesian product** with **Join**.



Heuristics based Query Optimization

- **Example:** **Restrict selection results** with **projection.**



Heuristics based Query Optimization

- **Example: Move PROJECT down:**

$$\sigma_{balance<2500}(\Pi_{balance}(account))$$

Rules for Transformation of Relational Expressions

- A **query** can be **expressed** in several **different ways**, with different **costs of evaluation.**
- Two **relational-algebra expressions** are said to be **equivalent** if, on every legal database instance, the **two expressions** generate the **same set** of **tuples.**
- **Equivalence Rules**
- An **equivalence rule** says that **expressions** of **two forms** are **equivalent.**
- Let $\theta_1\ \theta_2\ \theta_3$..................... Are **conditions**(predicates).
- **Rule 1**: Conjunctive selection operations can be **deconstructed** into a **sequence of individual selections**. This **transformation** is **referred** to as a cascade of **σ**.

$$\sigma_{\theta_1 \wedge \theta_2}(E) = \sigma_{\theta_1}(\sigma_{\theta_2}(E))$$

Rules for Transformation of Relational Expressions

- **Rule 2: Selection** operations are **commutative**.

$$\sigma_{\theta_1}(\sigma_{\theta_2}(E)) = \sigma_{\theta_2}(\sigma_{\theta_1}(E))$$

  - In practice, it is **better** and more **optimal** to **apply** that **selection first** which **yields** a **fewer number** of **tuples (restrictive selection).**
  - This **saves time** on our **outer selection.**

Rules for Transformation of Relational Expressions

- **Rule 3: Selections** on **Cartesian Products** can be re-written as **Theta Joins(Natural Join)**.

$$\sigma_\theta(E_1 \times E_2) = E_1 \bowtie_\theta E_2$$

- The **cross product operation** is known to be **very expensive.**
- This is because it **matches** each tuple of **E1** (total **m tuples**) with each tuple of **E2** (total **n tuples**).
- This **yields m*n entries**.

Rules for Transformation of Relational Expressions

- **Rule 4:** Theta Joins are **commutative.**

$$E_1 \bowtie_\theta E_2 = E_2 \bowtie_\theta E_1$$

- **Rule 5:** Join operations are **associative.**

$$(E_1 \bowtie E_2) \bowtie E_3 = E_1 \bowtie (E_2 \bowtie E_3)$$

- **Joins** are all **commutative** as well as **associative,** so one must **join those two tables first** which **yield less number of entries**, and then **apply** the **other join.**

Rules for Transformation of Relational Expressions

- Even though **join order** doesn't affect **query results** (because joins are commutative and associative), it can **drastically affect performance** because:
  - **Smaller intermediate results → faster joins**
  - **Joining selective (filtered) tables first → fewer rows to process next**
- **Example:** Let us consider **three tables:**

Customers (1 million rows)

| customer_id | name |

Orders (10 million rows)

| order_id | customer_id |

OrderDetails (100 million rows)

| order_detail_id | order_id | product_id |

Rules for Transformation of Relational Expressions

- Suppose we want to execute following **SQL statement:**
- *SELECT * FROM Customers c*
- *JOIN Orders o ON c.customer_id = o.customer_id*
- *JOIN OrderDetails od ON o.order_id = od.order_id WHERE c.customer_id = 123;*
- **Naive join order**
- *Orders ⋈ OrderDetails → very large intermediate result*
- *then filter to customer_id = 123*
- This **join** creates **massive intermediate data** before **filtering—it's slow** and **memory-intensive.**

Rules for Transformation of Relational Expressions

- **Better join order**:
- If **optimizer** instead **joins Customers** and **Orders first (with WHERE filter)**:
- *SELECT * FROM (SELECT * FROM Customers WHERE customer_id = 123)* ***c***
- *JOIN Orders* ***o*** *ON c.customer_id = o.customer_id*
- *JOIN OrderDetails* ***od*** *ON o.order_id = od.order_id;*
- Now Only **1 customer row** is selected,
- Only **that customer's orders** are **joined** (maybe **100 rows** instead of **10 million**)
- Then join those **100 orders** with **OrderDetails** (instead of 100 million rows)
- Thus **intermediate results** are **tiny**, making **query execution way faster.**
- Joining **smaller tables first** → **reduces** the **number of rows passed into subsequent joins.**

Rules for Transformation of Relational Expressions

- **Rule 6: Selection** operation can be **distributed**.

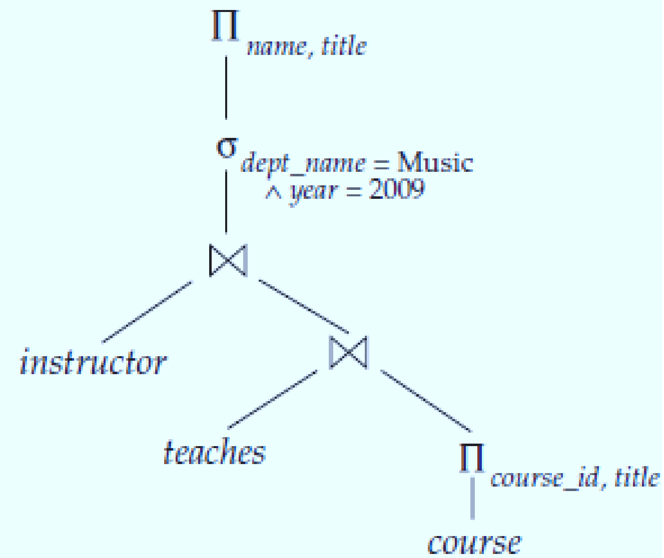
$$\sigma_{\theta_0}(E_1 \bowtie_\theta E_2) = (\sigma_{\theta_0}(E_1)) \bowtie_\theta E_2$$

$$\sigma_{\theta_1 \wedge \theta_2}(E_1 \bowtie_\theta E_2) = (\sigma_{\theta_1}(E_1)) \bowtie_\theta (\sigma_{\theta_2}(E_2))$$

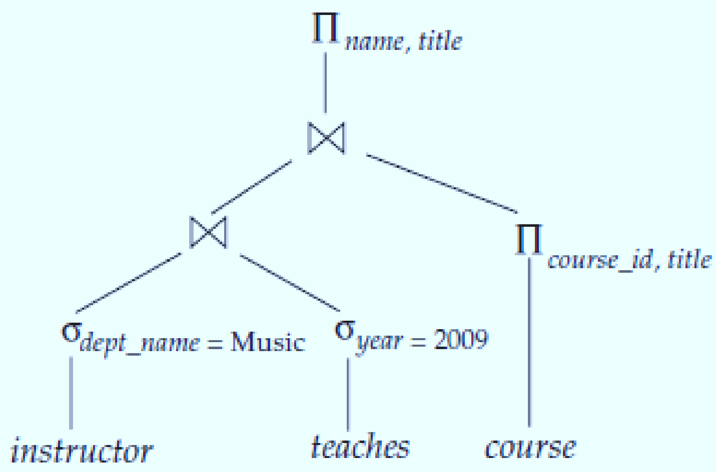
- **Rule 7: Projection distributes** over the **Theta Join.**

$$\Pi_{L_1 \cup L_2}(E_1 \bowtie_\theta E_2) = (\Pi_{L_1}(E_1)) \bowtie_\theta (\Pi_{L_2}(E_2))$$

Example



(a) Initial expression tree



(b) Tree after multiple transformations

Summary:Heuristics for Optimization

- ***Do operations*** *that* ***generate smaller tables*** *first.*
- Do **selection** as **early** as **possible.**
- Use **cascading, commutativity, and distributivity** to **move selection** as far **down** the **query tree** as possible.
- Use **associativity** to **rearrange relations** so the **selection operation** that will produce the **smallest table** will be executed **first.**
- Do **projection early.**
- Use **cascading, distributivity and commutativity** to **move the projection** as far **down** the **query tree** as possible.